

CS 159 Week 6:

Information retrieval & co-occurrence



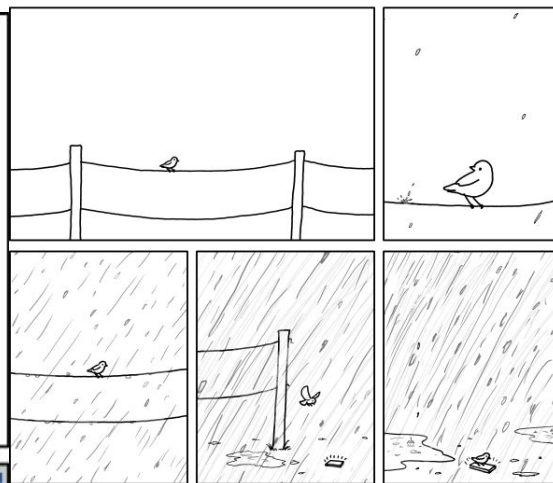
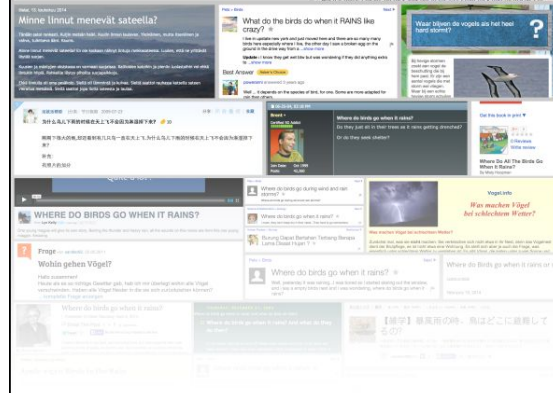
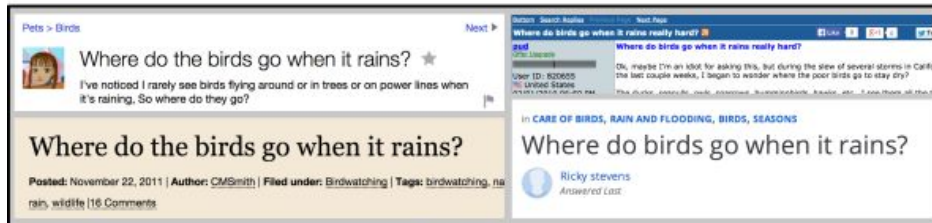
IT GIVES THE ANSWER, BUT ALSO SHOWS YOU AN ENDLESS TORRENT OF OTHER PEOPLE ASKING THE SAME QUESTION. PAGES AND PAGES OF THEM, ACROSS REGIONS AND CULTURES.



I LOVE THE IDEA THAT SOMEHOW THIS IS THE UNIVERSAL QUESTION, THE THING THAT UNITES US.

WHEN IT RAINS, WE WONDER WHERE THE BIRDS GO,

AND HOPE THEY'RE STAYING DRY.



(<https://xkcd.com/1434>)

Alert: *Potential* schedule change

Prof Chang *may* be out on the week of Nov 3 for an NLP conference (plans are still being decided)

The midterm may be moved up by a week (note: this might still happen anyway regardless of travel)

If I end up attending the conference, the plan for that week will be to (somehow) integrate it into the class! (since this is, after all, an NLP class)

Last week...

“Colorless green ideas sleep furiously.”



Colorless

green

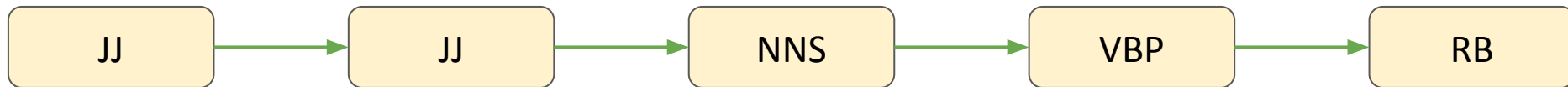
ideas

sleep

furiously

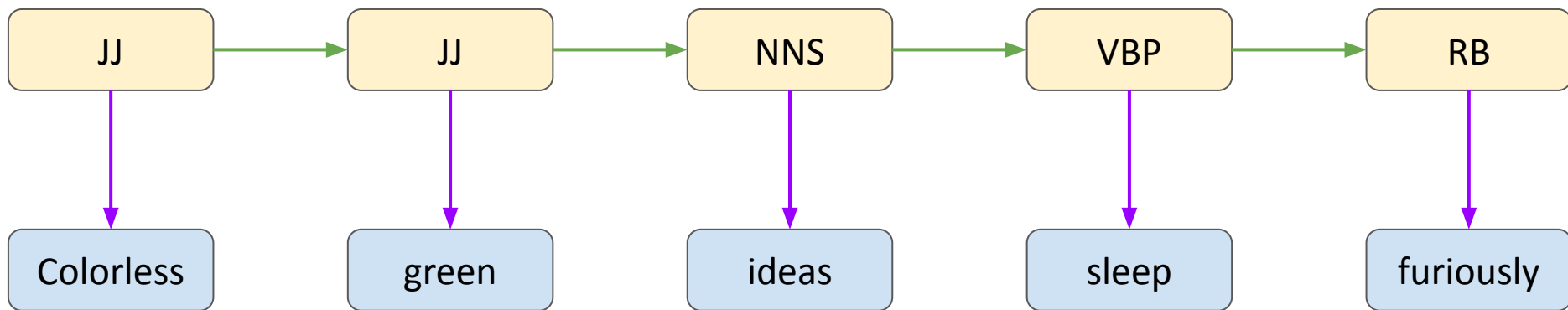
Last week...

“Colorless green ideas sleep furiously.”



Last week...

“Colorless green ideas sleep furiously.”

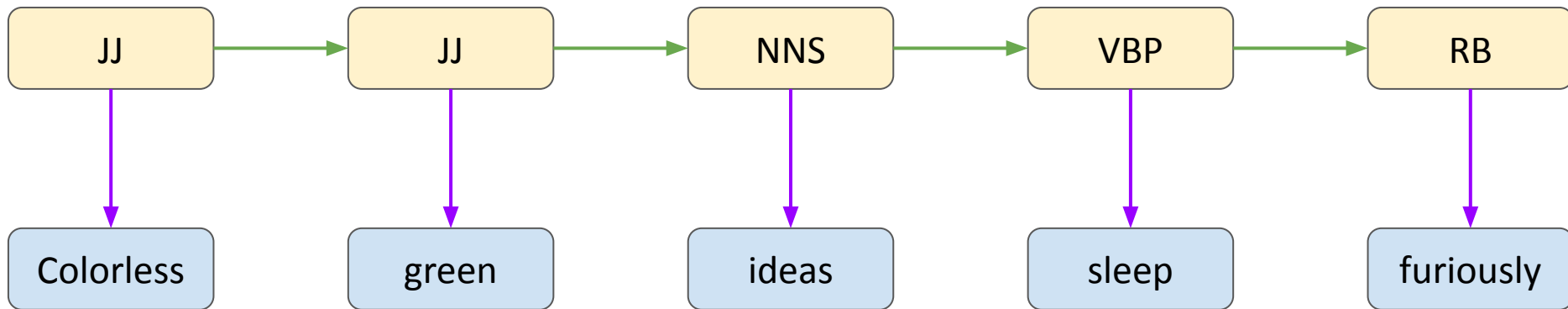


Last week...

“Colorless green ideas sleep furiously.”

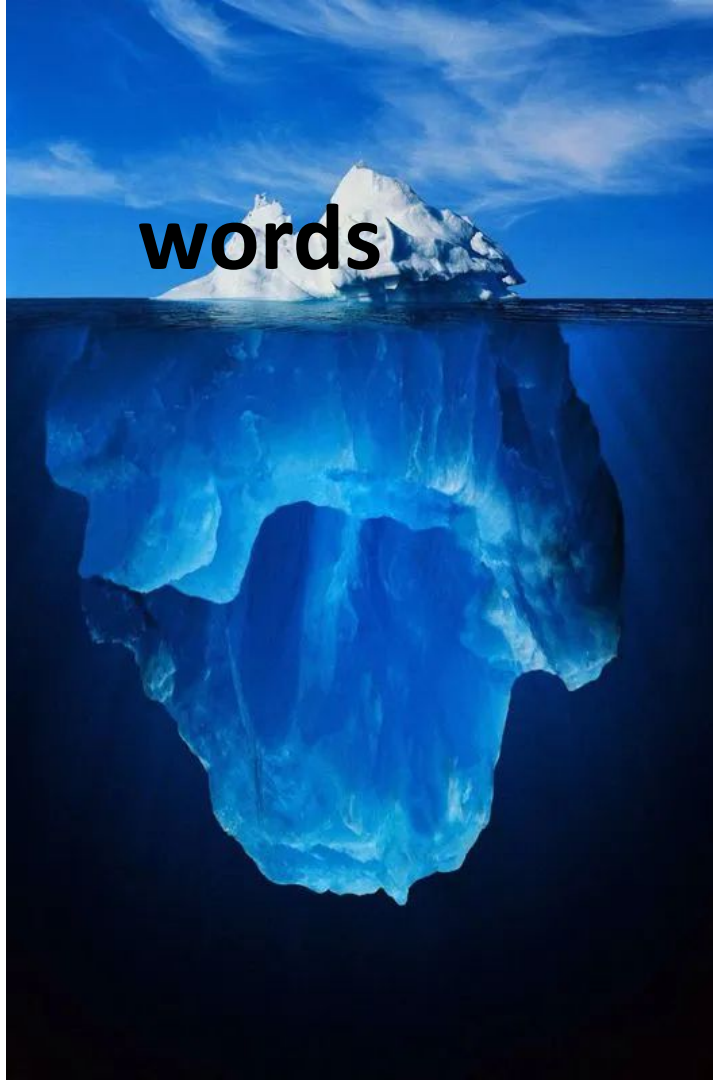


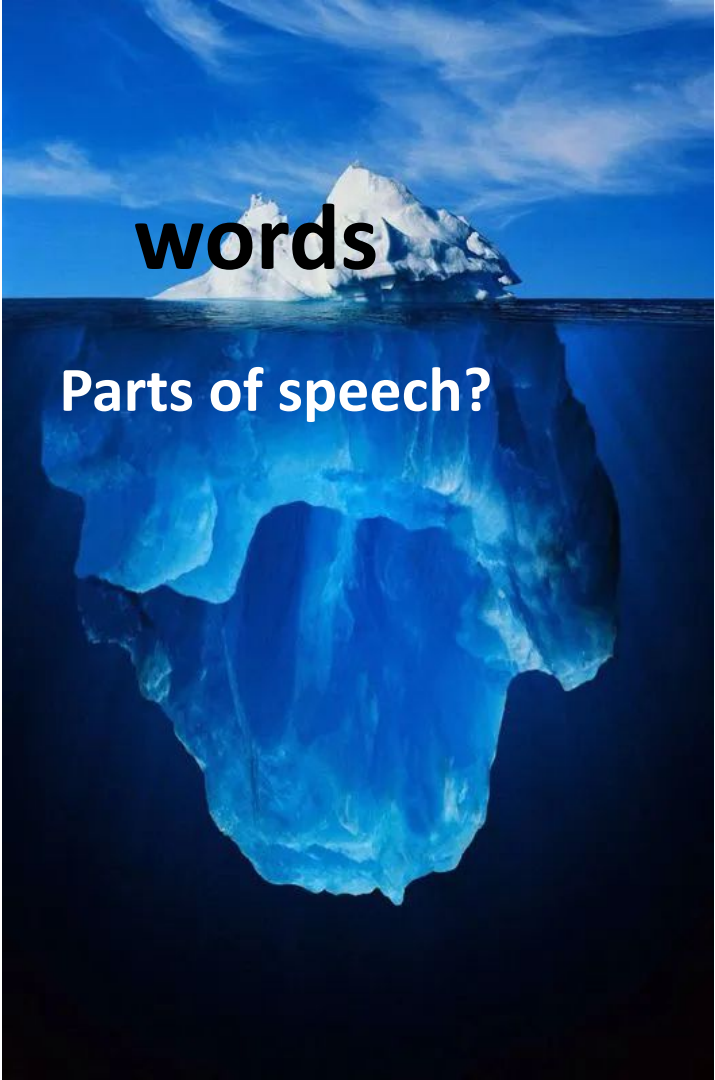
States



Observations

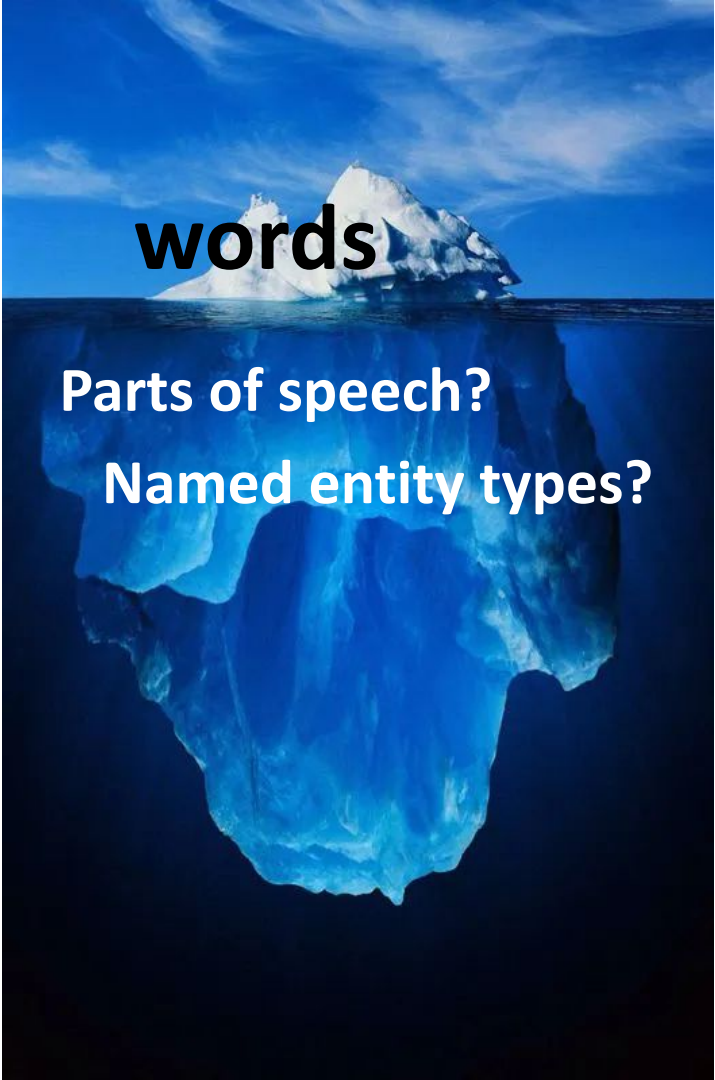
words



A photograph of an iceberg floating in the ocean. The tip of the iceberg is visible above the water line, while the much larger, more complex structure is submerged below. The sky is blue with some clouds, and the water is a deep blue. The text 'words' is overlaid on the visible tip, and 'Parts of speech?' is overlaid on the submerged part.

words

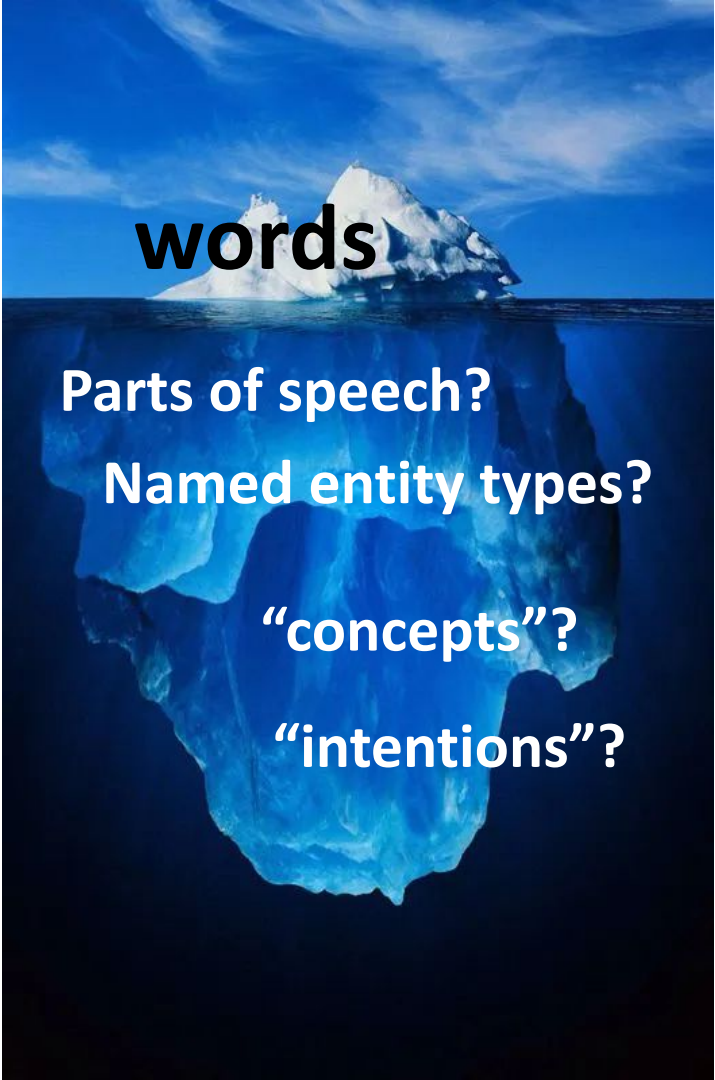
Parts of speech?

A photograph of an iceberg floating in the ocean. The tip of the iceberg is visible above the water line, while the much larger, more complex structure is submerged below. The sky is blue with some clouds, and the water is a deep blue.

words

Parts of speech?

Named entity types?

A vertical image of an iceberg floating in a dark blue ocean under a bright blue sky with wispy clouds. The iceberg's tip is above the water, while its much larger, jagged base is submerged. The text is overlaid on the image, with 'words' on the tip and the other questions on the submerged part.

words

Parts of speech?

Named entity types?

“concepts”?

“intentions”?

A journey from **words** to **language** to **meaning**

The past 3 weeks:

A journey from **words** to **language** to **meaning**

Today's agenda:

A journey from **words** to **language** to **meaning**

Real conversations you might hear around HMC campus



When is our midterm happening?

In about 4 more weeks!



Real conversations you might hear around HMC campus



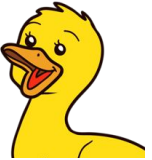
When is our midterm happening?

In about 4 more weeks!



What's your favorite Mudd building?

McGregor!



Real conversations you might hear around HMC campus



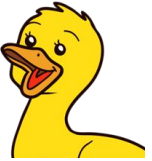
When is our midterm happening?

In about 4 more weeks!



What's your favorite Mudd building?

McGregor!



What's the Hoch exhibition serving today?

Sweet and sour pork!



Real conversations you might hear around HMC campus



When is our midterm happening?



A common pattern: all involve seeking **information**!

Information is requested through a **query**.

We seek an **answer** for each query.



today?

Sweet and sour pork!



A modest proposal

Instead of formally defining “meaning”, let’s be more indirect: can NLP be used to answer people’s *queries* (and thereby satisfy their information-seeking *intent*)?

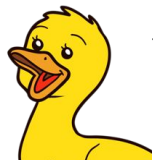
Problem: This might be tricky when all we have are words!

Key insight: for many queries, the answer probably already exists somewhere!

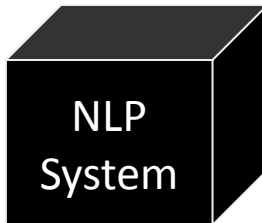
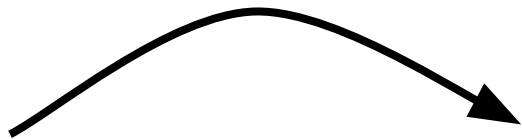


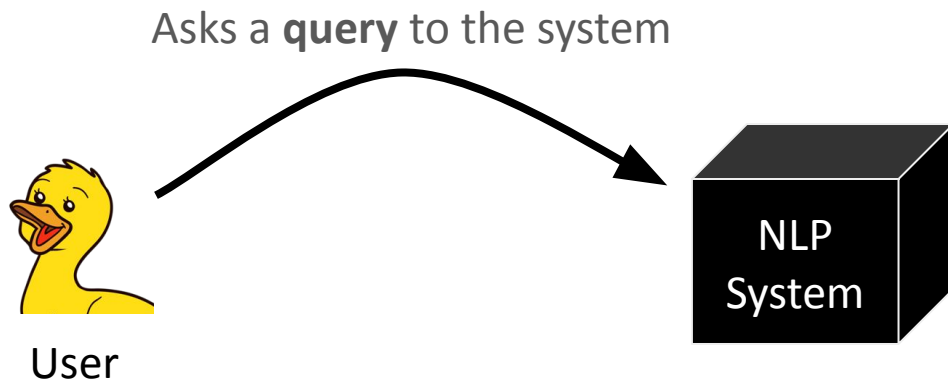
User

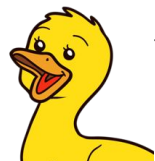
Asks a **query** to the system



User

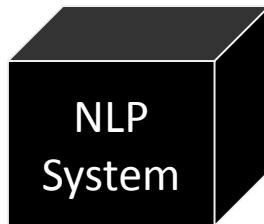






User

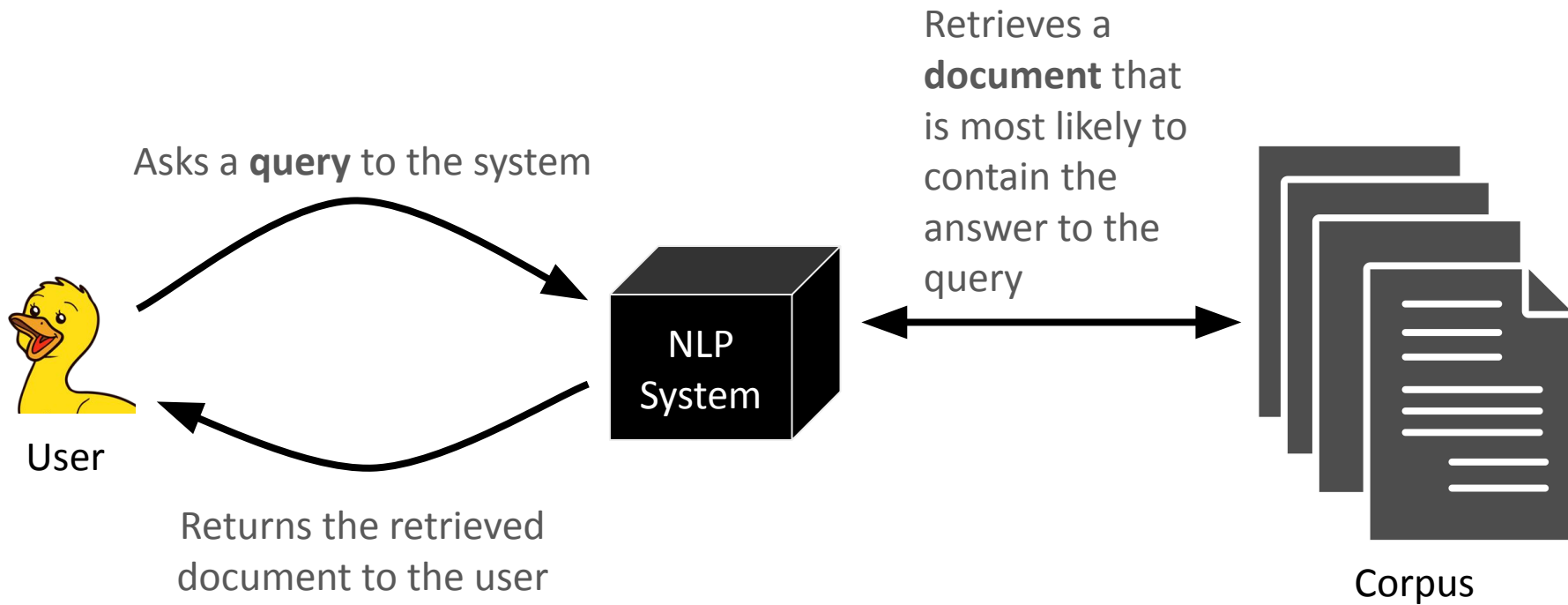
Asks a **query** to the system

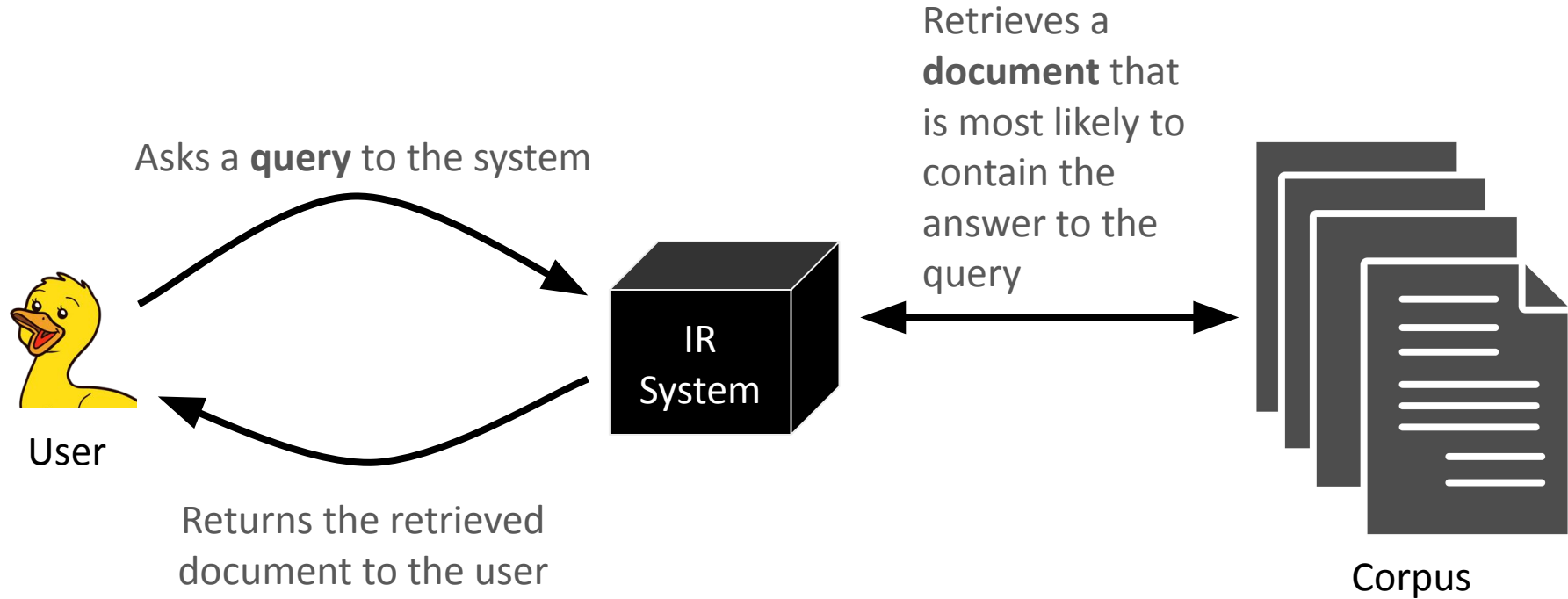


Retrieves a **document** that is most likely to contain the answer to the query



Corpus





Information Retrieval (IR)

Activity: acting out information retrieval!

- Each of you has been given one **query** and one *unrelated* **document**
- Goal: find which other person in the room has the answer to your query!
- Constraint: we only know about words, we don't know about "meaning", so...
 - You are only allowed to **ask questions** about **what words are in the other person's document**. You **cannot** just straight up ask your entire query to the other person, and you **cannot** ask "conceptual" questions like "is your document about animals?"
 - When answering questions, you can only directly answer what was asked, you can't use your own knowledge to "cheat" (e.g., if they ask "does your document talk about the president of HMC" and your document only says "Harriet Nembhard" the answer **should be no!**)
- You may find multiple candidate documents that might answer your question, in which case you should *rank* them (each document has a unique ID which you're allowed to ask for, you can use it to identify the candidates)

Answer key

Query ID	ID of document with answer
1	F
2	L
3	A
4	D
5	K
6	H
7	C
8	B

Query ID	ID of document with answer
9	J
10	M
11	G
12	P
13	N
14	E
15	Q

(Relative) word frequencies matter a lot for IR!

Intuitively, which of the following documents most likely contains an answer to “Where do birds go when it rains?”:

1. A document that says “bird” much more often than “rain”
2. A document that says “rain” much more often than “bird”
3. A document that says “rain” and “bird” about equally as often

Our choice of representation matters!

Thus far we have defined a “document” as a sequence of words:

$$d = w_1, w_2, w_3, \dots, w_k$$

Our choice of representation matters!

Thus far we have defined a “document” as a sequence of words:

$$d = w_1, w_2, w_3, \dots, w_k$$

But the truth is, this is actually just a **representation** of a document; a document is “actually” something more complex, like a web page, PDF file, etc.

Our choice of representation matters!

Thus far we have defined a “document” as a sequence of words:

$$d = w_1, w_2, w_3, \dots, w_k$$

But the truth is, this is actually just a **representation** of a document; a document is “actually” something more complex, like a web page, PDF file, etc.

This representation happens to be terrible for efficiently comparing relative word frequencies; can we think of a better one?

